NLP II Pop-Up Quiz: CNNs and RNNs

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1 Questions Only

Part A: CNNs for NLP

1. What is the main intuition behind using a CNN for text?
2. In a 1D text CNN, what does the convolutional filter slide over?
3. If each word in a sentence is represented by a d-dimensional embedding, what is the shape of the input sentence representation?
4. What does a filter width of 3 mean in a text CNN?
5. Why are CNNs often said to be good at detecting local patterns?
6. What kinds of linguistic patterns might a CNN detect in NLP tasks?
7. What is the role of the nonlinearity after convolution?
8. What does max-pooling do in a CNN for text?
9. Why is max-pooling useful for sentence classification?
10. What is the difference between using one filter and using many filters?
11. How does a CNN differ from a bag-of-words model?
12. Why can a CNN still lose some information about exact word order?

Part B: RNNs for NLP

13. What is the main intuition behind using an RNN for text?
14. What is the hidden state in an RNN?
15. Why are RNNs naturally suited to sequential data?
16. What information is passed from one time step to the next in a standard RNN?
17. What is the difference between a CNN and an RNN in how they process a sentence?

18. What is meant by saying that an RNN has a memory?
19. What problem is referred to as the vanishing gradient problem?
20. Why does the vanishing gradient problem matter for language?
21. How do LSTMs and GRUs improve on simple RNNs?
22. What is the difference between a unidirectional and a bidirectional RNN?
23. In what kind of NLP task would bidirectional context be especially useful?
24. Why might an RNN be preferred over a CNN for some language tasks?

Part C: Comparison and Conceptual Understanding

25. Which architecture is usually better at modeling local n-gram-like patterns: CNN or RNN?
26. Which architecture is usually more naturally aligned with incremental left-to-right processing?
27. Which is easier to parallelize across positions during training: CNN or standard RNN?
28. If the task is sentence classification, why might a CNN work very well?
29. If the task requires tracking dependencies over many words, why might an RNN or LSTM be a better fit?
30. Give one advantage of CNNs over RNNs.
31. Give one advantage of RNNs over CNNs.
32. What is one limitation shared by both CNNs and RNNs compared with transformers?

2 Questions and Answers

Part A: CNNs for NLP

1. **What is the main intuition behind using a CNN for text?**
   A CNN detects local patterns in a sequence, such as short phrases or neighboring word configurations, and uses them as features for prediction.

2. **In a 1D text CNN, what does the convolutional filter slide over?**
   It slides over the sequence of word embeddings, moving across token positions in the sentence.

3. **If each word in a sentence is represented by a d-dimensional embedding, what is the shape of the input sentence representation?**
   For a sentence of length n, the input is typically an $n \times d$ matrix.

4. **What does a filter width of 3 mean in a text CNN?**
   It means the filter looks at windows of 3 consecutive words at a time.

5. **Why are CNNs often said to be good at detecting local patterns?**
   Because each filter examines only a small local region of the input and learns to respond strongly when a useful local pattern appears.

6. **What kinds of linguistic patterns might a CNN detect in NLP tasks?**
   It might detect negation patterns, local syntactic fragments, sentiment-bearing phrases, or short collocations.

7. **What is the role of the nonlinearity after convolution?**
   The nonlinearity allows the model to learn more expressive, non-linear features rather than only linear combinations.

8. **What does max-pooling do in a CNN for text?**
   It selects the largest activation from a feature map, keeping the strongest detected instance of a pattern.

9. **Why is max-pooling useful for sentence classification?**
   Because the task may depend on whether a useful pattern appears anywhere in the sentence, not exactly where it appears.

10. **What is the difference between using one filter and using many filters?**
    One filter learns one kind of pattern; many filters learn many different kinds of patterns.

11. **How does a CNN differ from a bag-of-words model?**
    A CNN is sensitive to local order and adjacency, whereas bag-of-words ignores word order entirely.

12. **Why can a CNN still lose some information about exact word order?**
    Because after pooling, the model often preserves that a feature occurred, but not always its exact position in the sequence.

Part B: RNNs for NLP

13. **What is the main intuition behind using an RNN for text?**
    An RNN processes a sentence one word at a time, updating a running state that summarizes what has been seen so far.

14. **What is the hidden state in an RNN?**
    The hidden state is the internal vector representation that carries contextual information from earlier time steps.

15. **Why are RNNs naturally suited to sequential data?**
    Because they are designed to process inputs in order, passing information from one step to the next.

16. **What information is passed from one time step to the next in a standard RNN?**
    The hidden state is passed forward and updated with the current input.

17. **What is the difference between a CNN and an RNN in how they process a sentence?**
    A CNN examines local windows in parallel, while an RNN processes the sentence step by step in sequence.

18. **What is meant by saying that an RNN has a memory?**
    It means the hidden state can retain information from earlier words and use it when processing later words.

19. **What problem is referred to as the vanishing gradient problem?**
    During training, gradients can become extremely small as they are propagated backward through many time steps, making it hard to learn long-range dependencies.

20. **Why does the vanishing gradient problem matter for language?**
    Because important contextual information may come from words much earlier in the sentence, and a simple RNN may fail to learn such long-distance effects.

21. **How do LSTMs and GRUs improve on simple RNNs?**
    They introduce gating mechanisms that help preserve, update, and forget information more effectively over long sequences.

22. **What is the difference between a unidirectional and a bidirectional RNN?**
    A unidirectional RNN reads in one direction only; a bidirectional RNN reads both left-to-right and right-to-left and combines the two contexts.

23. **In what kind of NLP task would bidirectional context be especially useful?**
    Tasks like tagging, parsing, or word sense disambiguation, where both left and right context help interpret a token.

24. **Why might an RNN be preferred over a CNN for some language tasks?**
    Because an RNN is better suited to modeling sequential context and dependencies that unfold over time.

Part C: Comparison and Conceptual Understanding

25. **Which architecture is usually better at modeling local n-gram-like patterns: CNN or RNN?**
    CNN.

26. **Which architecture is usually more naturally aligned with incremental left-to-right processing?**
    RNN.

27. **Which is easier to parallelize across positions during training: CNN or standard RNN?**
    CNN.

28. **If the task is sentence classification, why might a CNN work very well?**
    Because sentence classification often depends on the presence of informative local features anywhere in the sentence, which CNNs detect efficiently.

29. **If the task requires tracking dependencies over many words, why might an RNN or LSTM be a better fit?**
    Because it maintains a contextual state across the sequence and can, especially in gated variants, better capture longer-range dependencies.

30. **Give one advantage of CNNs over RNNs.**
    CNNs are usually faster and more parallelizable, and they are very effective at learning local patterns.

31. **Give one advantage of RNNs over CNNs.**
    RNNs more directly model sequential structure and evolving context through the sentence.

32. **What is one limitation shared by both CNNs and RNNs compared with transformers?**
    Both generally have more difficulty capturing very long-range dependencies and global interactions as flexibly as transformers do.